

QP Code: D133983		Total Pages: 1	Name:
			Register No.
THIRD SEMESTER UG DEGREE EXAMINATION, NOVEMBER 2025			
(CUFYUGP)			
CSC3MN206 - PROGRAMMING USING PYTHON			
2024 Admission onwards			
Maximum Time: 2 Hours			Maximum Marks: 70
Section A			
All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)			
1	What is an algorithm? Give an example.		
2	Explain modularity in problem solving.		
3	Define a variable and give examples of valid identifiers.		
4	What is a conditional statement?		
5	What is fancy indexing in NumPy?		
6	Explain the difference between min() and argmin() functions in NumPy.		
7	What is the purpose of the return statement in a function?		
8	Define and give one example of a dictionary in Python.		
9	What is meant by multiple assignment?		
10	Write the syntax and example of a nested conditional.		
Section B			
All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)			
11	Describe step-by-step the process of converting a problem into a flowchart.		
12	Explain order of precedence and associativity in Python expressions.		
13	Explain how aggregation functions work in NumPy with examples.		
14	Discuss the concept of logical operators and their role in Boolean expressions.		
15	Describe the use of loops with examples for generating multiplication tables.		
16	Explain the concept of comparisons and masks in NumPy.		
17	Write a Python function to count the vowels in a given string.		
18	Explain string traversal and comparison operations with examples.		
Section C			
Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)			
19	Design algorithms and flowcharts for calculating factorial of a number and finding the largest among three numbers.		
20	Write a Python program using NumPy to perform addition, subtraction, and multiplication of two matrices.		